~/paulynomial

paul kiernan

staff+ infrastructure engineer — search, retrieval & streaming systems

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summary

Fifteen years building search and streaming infrastructure at scale. At Reddit (500M+ monthly users), led search indexing for a multi-billion-document Solr/Lucene index, shipped comment search (7.5M+ new comments a day, searchable within 60 seconds), and designed the 5M+ messages/sec Kafka platform other teams built on; the same systems later served Reddit's LLM products. Founding engineer, then CTO, at Phaselaw: built an AI redaction platform from zero to SOC 2 Type II and 10M+ documents for the NHS and others.

experience

Sabbatical — San Francisco, CA — Nov 2025 – Present

- Planned break after Phaselaw: time with family, and independent study of what retrieval infrastructure looks like when agents, not humans, are the query interface.

Phaselaw — Remote (San Francisco, CA) — Jul 2023 – Nov 2025

Chief Technology Officer — Oct 2024 – Nov 2025

- Led engineering for an AI-powered document review and redaction platform handling highly sensitive legal and personal data: 10M+ documents redacted for the NHS, the UK Labour Party, and the Raspberry Pi Foundation.
- Built and led the engineering organization; owned hiring, engineering culture, and technical due diligence with investors and partners.
- Took the platform through SOC 2 Type II: CI/CD, automated testing, OpenTelemetry observability, and multi-region Terraform on AWS (US, EU, and UK for data residency); stack: LLM-based PII identification (Claude on AWS Bedrock), Temporal orchestration, multi-tenant Kubernetes with ArgoCD GitOps.

Founding Backend Engineer — Jul 2023 – Oct 2024

- Designed and built the core backend from zero (data models, API layer, authn/authz, service architecture) across multiple global regions; made the foundational technology choices, favoring boring, proven technology so a small team could ship and operate a global platform.

Reddit — San Francisco, CA — Jul 2018 – Jul 2023

Staff Software Engineer — Tech Lead, Search Indexing — Nov 2019 – Jul 2023

- Led the search indexing team: managed 3 engineers and mentored 3 more while staying hands-on, and owned the end-to-end design of the Scala/Apache Flink pipeline keeping a multi-billion-document Solr/Lucene index fresh against millions of events per second.
- Shipped comment search, a capability Reddit had lacked for a decade after post search, by architecting the Kafka infrastructure and consumer DAG that indexes 7.5M+ new comments/day within 60 seconds of posting.
- The same indexing and retrieval systems later served all of Reddit's LLM applications.
- Designed the Kafka platform that finally scaled for Reddit: multiple self-managed clusters with tuned partitioning and replication and a new consumer-processor pattern; a 5M+ messages/sec backbone adopted by the analytics, data warehousing, and subreddit notification teams.
- Led search incident response and materially improved Reddit Search uptime.

Senior Software Engineer II, Data Infrastructure — Jul 2018 – Nov 2019

- Replaced poll- and cron-based batch jobs with event-driven analytics and Kafka-to-BigQuery warehouse ingestion pipelines, cutting downstream latency from hours to seconds.

Chartbeat — New York, NY — Aug 2014 – Jul 2018

Head of Backend Engineering — Oct 2016 – Jul 2018

- Led 8 backend engineers across 4 product teams; earlier led the R&D team (Backend Team Lead) and presented experimental products to the board.
- Built MrRoboto, a replicated C++ key-value store sustaining 500k+ writes/sec (LMDB copy-on-write B-tree, chain replication); Miata, a real-time streaming analytics framework on Kafka, ZeroMQ, and Python; and the probabilistic-counting backend for the "so far today" API. Open-sourced Parselmouth.

Moat (acquired by Oracle) — New York, NY — Jun 2012 – Aug 2014

Software Developer

- Early engineer on Moat Pro, an enterprise ad-intelligence platform; built the geo-distributed web-crawler network, ETL pipeline, and reporting API that detected, captured, and indexed online advertisements.

skills

languages	Python, Scala, C++, Java, C, Clojure, SQL, Shell
infrastructure	Kafka, Apache Flink, Solr/Lucene, BigQuery, Temporal, ZeroMQ, AWS (multi-region), Kubernetes, Terraform, ArgoCD, OpenTelemetry, HAProxy
ai / llm	LLM extraction pipelines (Claude on AWS Bedrock), retrieval infrastructure for LLM applications

education & patents

education	**Cornell University**, BS in Electrical & Computer Engineering, 2012. McMullen Dean Scholar; Goldfarb Tradition Fellow.
patent	**US 10,068,250**: "System and method for measuring mobile advertising and content by simulating mobile-device usage."